

Laser welding of fused silica glass with sapphire using a non-stoichiometric, fresnoitic $\text{Ba}_2\text{TiSi}_2\text{O}_8 \cdot 3 \text{SiO}_2$ thin film as an absorber

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ABSTRACT

Laser welding of dissimilar materials is challenging, due to their difference in coefficients of thermal expansion (CTE). In this work, fused silica-to-sapphire joints were achieved by employment of a ns laser focused in the intermediate Si-enriched fresnoitic glass thin film sealant. The microstructure of the bonded interphase was analyzed down to the nanometer scale and related to the laser parameters used. The crystallization of fresnoite in the glass sealant upon laser process leads to an intense blue emission intensity under UV excitation. This crystallization is favored in the interphase with the silica glass substrate, rather than in the border with the sapphire. The formation of SiO_2 particles was confirmed, as well. The bond quality was evaluated by scanning acoustic microscopy (SAM). The substrates remain bonded even after heat treatment at 100 °C for 30 min, despite the large CTE difference between both substrates.

1. Introduction

Laser joining of dissimilar materials is a topic of great importance in a wide range of fields, including electromechanical, electronic, and medical applications. Since the heat-affected zone imposed by laser interaction with matter is typically in the micrometer range, laser welding allows joining dissimilar materials, minimizing the impact of thermal expansion differences [1].

Welding opaque to transparent materials by ultrashort pulsed lasers (ps, fs) has been extensively reported, including glass-silicon, metal-glass, [2,3] and even sapphire-to-stainless steel [4] and copper-to-glass by nanosecond pulsed lasers [5]. However, the challenge of welding two different transparent materials is still an issue in great development. This field finds several interesting industrial applications in optics, optoelectronics, microelectromechanical systems (MEMS), microfluidics, in medical devices, etc. [6,7]. Bonding methods like gluing with adhesives, optical contacting and heat treatment offer weak bond strength and poor chemical durability, or lead to the formation of thermal stresses in the case of heat treatment. Thus, laser welding appears to be a good candidate. In comparison to CO_2 -lasers, visible to near-infrared pulsed lasers can be focused in the interior of the glass substrate, without affecting the glass surface. In our recent publication, two borosilicate glasses were joined by applying a titanium ultrathin film as sealant in between, and even without any intermediate sealant layer, by using a ns pulsed laser [8]. Among the glass-to-glass micro-welding

works, the publications of Miyamoto et al. are worth mentioning. They described the glass-to-glass welding by employing an ultrashort pulse lasers, including theoretical approaches to the internal modification of glasses by fs and ps pulsed lasers [9–11].

Works dedicated to the laser welding of two dissimilar transparent materials are found in literature, as well. Good examples are the works of Watanabe et al. [12,13] (and references therein) who reported the joining of a borosilicate glass to fused silica through fs-pulsed laser irradiation. Other examples include fused silica to BK7 glass assemblies by employing a fs pulsed laser [14].

Glass sealants have become promising materials for joining similar or dissimilar materials [15]. They crystallize into a glass-ceramic during the sealing process. Glass-ceramics present better mechanical properties than glasses and offer the possibility to tune their coefficient of thermal expansion (CTE) by controlling the crystalline fraction of a certain CTE. This is particularly important for avoiding stresses during bonding.

Laser welding of transparent materials using glass sealants has been performed by using continuous lasers at 10.6 μm wavelength, since the glasses present high light absorption at this wavelength. Here, the laser beam is coupled either directly into the glass sealant by locating the sandwich substrate-sealant-substrate transversal, or indirectly, where the laser beam is coupled into the metal substrate and the glass sealant melts as a result of heat conduction in the metal [16]. Ultra-short pulsed lasers at 1064 nm at high frequencies have been used, as well [8,9].

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Among transparent materials, sapphire is widely used in the field of optical components and micromechanical devices [17,18]. Sapphire possesses high mechanical stability and chemical durability, is heat resistant, has a high ablation threshold, and exhibits optical transparency over a large spectral range [19]. According to the Mohs scale, sapphire is the second hardest transparent crystalline material, after diamond. Its applications include windows for high-pressure optical cells and high-pressure devices [20]. Sapphire has been bonded to other materials using different techniques and using various solder materials [21–23].

Other important transparent material is fused silica glass. It possesses high optical transmission, biocompatibility, chemical and mechanical durability, etc. Fused silica is widely used in several applications within MEMS, microfluidic devices, resonators, etc. [24–26]. The transparency of fused silica is very useful for visual inspection of the bonded area. Two fused silica substrates have been joined by fs-lasers (1 μm wavelength) without any absorber in between [27]. Welding using a 10.6 μm CO_2 laser has been also reported, due to the massive absorption of fused silica at this wavelength [28].

In our previous publications [29–31], we applied a fresnoitic glass ($2\text{BaO}\cdot\text{TiO}_2\cdot 2\text{SiO}_2$, BTS) thin film as an absorber for the laser light, acting as a solder between two transparent substrates. In this way, silica glass-to-silica glass and sapphire-to-sapphire joints were formed. BTS exhibits fluorescence, it is pyro-, piezo-, and ferroelectric [32], and possesses non-linear optical properties [33]. Ti^{4+} ions hosted in a square-pyramidal TiO_5 are responsible for blue fluorescence when irradiated in the UV range [34]. In our previous contributions [29–31], we reported the damage of bond interface with the upper substrate by the laser irradiation. Upon UV excitation the solder strongly emits in the blue spectral range. This fluorescence was attributed to fresnoite crystals, formed during the laser welding process [30].

In the present work, a SiO_2 -enriched fresnoite glass thin film is investigated as glass solder between a fused silica substrate and a sapphire substrate as bonding partner. By increasing the SiO_2 content, the CTE of the thin film should decrease, making it more compatible to the CTE of the joined substrates.

The novelty of this work lies by one hand, in the use of a ns pulsed laser to weld two transparent materials, like sapphire and silica glass. The use of a nanosecond pulsed laser would represent an advantage against ultra-short pulsed lasers, since the formers are more stable and can be fabricated at lower costs than the laterers. By the other hand and for the best of our knowledge, it is the first time that silica glass and sapphire are laser-welded through a glass thin film as a solder.

2. Experimental

2.1. Film growth

A Fresnoite glass (BTS) with an excess of SiO_2 of nominal composition $2\text{BaO}\cdot\text{TiO}_2\cdot 5\text{SiO}_2$, from now on as $\text{BTS}\cdot 3\text{SiO}_2$, was prepared from reagent-grade BaCO_3 , TiO_2 , and Si_2O by using the conventional melt-quenching technique. Then, the mixture was melted in a platinum crucible at 1600 $^\circ\text{C}$ for 1 h in air. The obtained glass ($\text{BTS}\cdot 3\text{SiO}_2$) was used as target to coat on 2 in. diameter polished c-plane sapphire wafers (thickness 430 μm) by off-axis pulsed laser deposition (PLD) [35]. 120,000 laser pulses from a KrF excimer laser (wavelength 248 nm) were applied with a pulse frequency of 20 Hz. The $\text{BTS}\cdot 3\text{SiO}_2$ thin films were grown on the substrate kept at 180 $^\circ\text{C}$ and with 0.001 mbar oxygen partial pressure.

$\text{BTS}\cdot 3\text{SiO}_2$ -coated sapphire wafers, as well as uncoated fused silica wafers of 500 μm thickness, were cut into $10\times 10\text{ mm}^2$ pieces. Fig. 1 schematically illustrates the employed set-up. The uncoated silica glass substrate was then placed over the $\text{BTS}\cdot 3\text{SiO}_2$ -coated sapphire substrate leaving the $\text{BTS}\cdot 3\text{SiO}_2$ thin film acting as glass solder (Fig. 1). The set-up for the laser bonding is the same shown previously in [29,31], in which a custom-made arrangement fixture was used in order to bring both substrates in close proximity.

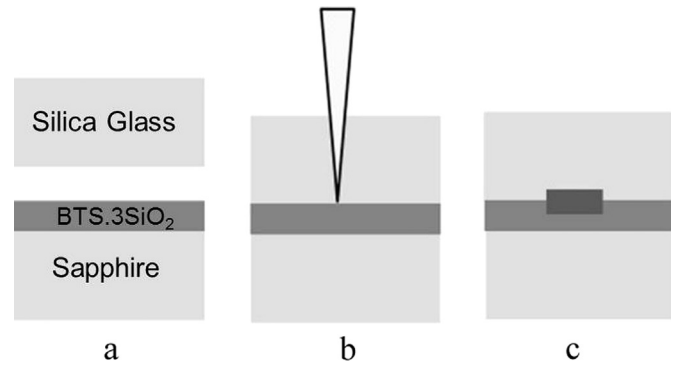


Fig. 1. Set-up. (a) The uncoated fused silica substrate is placed over the $\text{BTS}\cdot 3\text{SiO}_2$ -coated sapphire substrate. For clarity, the used sample holder is not represented. (b) The laser goes through the uncoated silica substrate and is focused at the $\text{BTS}\cdot 3\text{SiO}_2$ interface. (c) Patterns of horizontal lines were irradiated.

2.2. Laser bonding

The irradiation source was a pulsed 5 nanosecond Nd: YAG laser (Xiton Photonics Laser) implemented into a microSTRUCT C laser micromachining workstation by 3D-Micromac AG, operating at 532 nm. A galvanometer scanner unit was employed, and the laser beam moves over the workpiece that remains fixed. An objective of 255 mm focal length was employed and the beam diameter formed is 23 μm . The laser beam goes through the silica glass substrate and is precisely focused at the interface between both substrates (Fig. 1b-c), since fused silica is practically transparent to the wavelengths used. The focus position changes when crossing a transparent material. When focusing from air ($n_1=1$) into a material of refractive index n_2 , refraction at the interface moves the focus from a value z to z' , according with the equation: $z'=n_2\cdot z$ [36]. Thus, the focal point in the intermediate fresnoite thin film was calculated taking into account the thickness of the upper silica glass, $z'=500\text{ }\mu\text{m}$, and its refractive index n_2 at the applied wavelength of 532 nm, 1.4607.

A wobble approach of the laser beam was used, that is, a circular motion of the laser beam [37] is superimposed to a movement along a given trajectory. By setting the transversal and longitudinal amplitudes of the loops and the frequency of the wobble movement (number of loops per second), the desired wobble trajectory is obtained. The wobble parameters were optimized to: Transversal amplitude: 0.08 mm, longitudinal amplitude: 0.08 mm and wobble frequency: 50 Hz (50 loops/s). After a laser parameter optimization, the laser repetition rate was fixed to 35 kHz and the laser fluence and pulse energy to 8.25 J/cm^2 and 34.3 μJ , respectively. Scan speed and the number of passes per line were optimized through series of experiments according with Table 1. The corresponding pulse overlap was calculated for each scan speed. Each line was irradiated several times (number of passes/line in Table 1) in a bidirectional mode, to favor the heat accumulation. Patterns consisting of 10 horizontal lines forming a block of lines were irradiated in the same sample, named from A to E in Table 1. The distance between irradiated lines was optimized to 100 μm to ensure an approximate 50% overlap of the lines and thus, a fully bonded area, based on our previous work [31].

Table 1
Laser parameters used for the tests shown in Fig. 2.

Block	Lineal scan speed (mm/s)	Pulse overlap (%)	Number of Passes/ line
A	2	99.75	14
B	5	99.36	14
C	2	99.75	20
D	5	99.36	20
E	10	98.73	40

The parameters of the block C were selected to develop additional samples. Squared patterns of $5 \times 5 \text{ mm}^2$ of horizontal lines with $100 \mu\text{m}$ between lines were irradiated under the C block conditions. Moreover, a further welded pattern was irradiated, consisting of two blocks of 20 horizontal lines. The first block was irradiated at 2 mm/s and 20 passes/line, (like block C) while the second block was performed at 5 mm/s and 20 passes/line as well (block D).

2.3. Scanning acoustic microscopy (SAM)

The bonded area of the test parameter sample joints was investigated by scanning acoustic microscopy (SAM). SAM is a non-destructive technique used for characterizing bonded wafers. The acoustic microscope employed here was an SAM400 (PVA TePla Analytical Systems GmbH, Westhausen, Germany) in combination with an ultrasonic transducer (Siegert TFT, Hermsdorf, Germany) with 175 MHz center frequency and a focal length of 4 mm in water. For acoustic coupling, the samples were submerged in de-ionized and degassed water at 21 °C. The coupling medium was required for matching the acoustic impedances of the sample surface and the front of the acoustic lens. To improve acoustical imaging within solid samples, the focus of the ultrasonic transducer was positioned at the bonding interface in between the two substrates. The acoustic micrographs were computed offline from the recorded echo signals. A sliding window algorithm was used to determine the correct interface echo time delay and for optimization purposes two dimensional image filtering was performed. The contrast of the acoustic images was also adjusted and binarized to derive the effective bonded area. All signal and data processing was performed using custom-made analysis software based on MATLAB.

2.4. Optical characterization

The bonded samples were examined with an optical microscope (Leica DM RXE-650H).

Fluorescence maps were recorded in the same optical microscope by locating an UV Lamp Camag (254 nm excitation wavelength with excitation density of 1.2 mW/cm^2) in front of the bonded sample. The excitation light was separated from the emitted fluorescence using the following filters: a BP 340–380 filter for excitation and a LP 425 filter for emission. The fluorescence was collected with 50 s of acquisition time using a Leica digital camera as detector system incorporated in the microscope.

2.5. Structural characterization by scanning electron microscopy (SEM) and energy dispersive X-ray spectroscopy (EDXS)

The microstructure of the bond interface of one of the samples was characterized by SEM in both, silica glass and sapphire substrates after its mechanical separation.

A planar and cross-sectional sample of one of the bonded specimens in which two block of lines were irradiated was prepared using an argon ion beam (Gatan Ilion⁺™ II). A Carl Zeiss Supra 55VP SEM microscope was employed, with secondary electron (SE) detector and angle-selective backscattered electron (AsB) detector. EDXS analyses of Ba (L line, 5247 eV), Ti (K-line, 4966 eV) and Si (K line, 1839 eV) were recorded at 20 kV in the same microscope.

2.6. (Scanning) Transmission electron microscopy ([S]TEM)

The cross-section of the stack consisting of both sapphire and quartz substrates joined through the BTS-3 SiO₂ thin film was examined by TEM/STEM. The bond was joined applying the laser parameters C (Table 1). In order to obtain electron-transparent sections, the stack was cut in the middle, parallel to the laser irradiated lines. Then, plane-parallel sections were sliced perpendicular to the

laser writing direction, ground plane-parallel and a wedge-polishing route was developed: using a specific tripod sample holder in combination with a multi-functional grinding and polishing tool (MultiPrep, Allied Company), a very thin, electron-beam transparent wedge was generated, by polishing it under a defined, very small angle ($\approx 1.6^\circ$). After mounting on a metal half ring, the sample was subsequently subjected to double-sided, ion-beam polishing using low-energy (2.5 keV) Ar⁺ ions under a small angle of incidence (6°) until the potential damage slabs (due to mechanical treatment) were removed and the samples became electron transparent (precision ion-polishing system, PIPS, Gatan, Inc.). In order to avoid charging of the non-conducting cross-section samples under electron irradiation, selective carbon coating was accomplished [38]. Bright field images in TEM mode were recorded with a FEI Tecnai G2 F20 microscope operating at 200 kV. Scanning TEM (STEM) and EDX line scan analysis of Al (K line), Ba (L line), Ti (K line) and Si (K line) were done in the same TEM microscope.

3. Results

Fig. 2a shows an overview of the joined sample where five blocks were irradiated according to parameters A to E (Table 1). The laser beam was focused through the upper fused silica substrate. The free surfaces of the transparent substrates are free of damages and only the interface between them shows the signs of the interaction with the laser light. Irradiated horizontal lines are observed in each block, all of them exhibit a milky color in comparison with the non-irradiated areas. Under UV exposure (254 nm), the irradiated blocks emit in the blue. This is a first indication of the glass solder modification upon laser irradiation. A different intensity is observed for each block, as shown in Fig. 2b, with the highest intensity being registered for block C, which

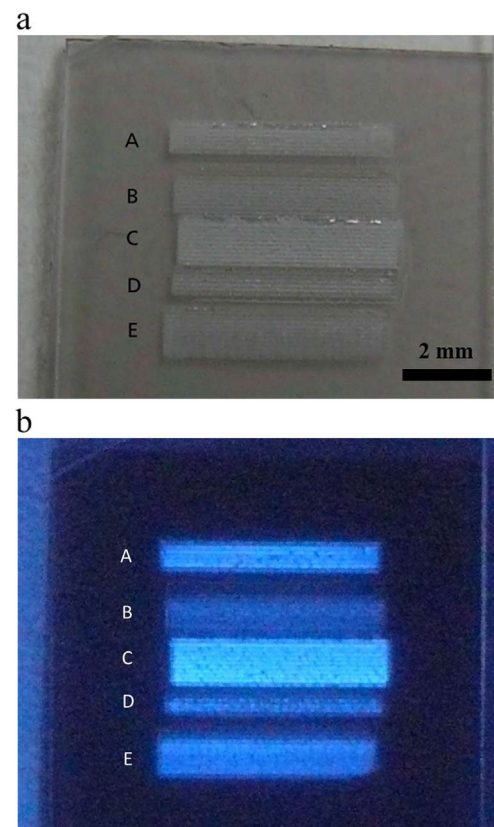


Fig. 2. (a) Overview top-view, optical micrograph of the joined fused silica substrate and the sapphire substrate through the BTS-3 SiO₂ thin film. (b) Optical top-view micrograph under 254 nm excitation. The letters indicate the laser parameters in Table 1. Fused silica substrate is upwards in both micrographs.

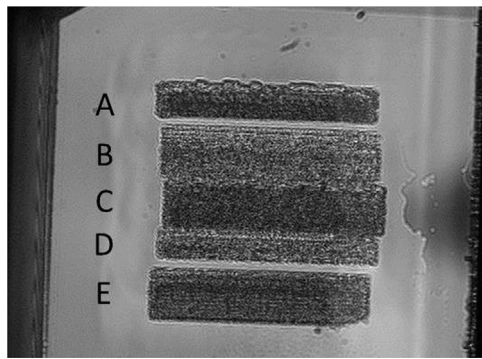


Fig. 3. SAM micrograph of the joined fused silica substrate and the sapphire substrate through the BTS-3 SiO₂ thin film. The letters indicate the laser parameters in Table 1.

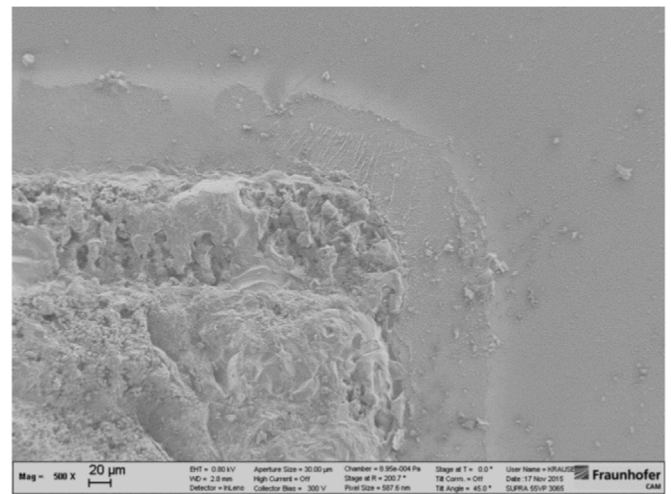
was irradiated under the slowest scan speed of 2 mm/s, and thus, a higher pulse overlap, and 20 passes per line (Table 1).

A good bond quality is characterized by the absence of cracks and a no gaps at the joint interfaces. In order to inspect the bonded areas and to detect less perfectly bonded regions, the very same sample was examined by SAM [39,40]. SAM micrographs are digitalized in the way that non-bonded areas, in which the water used as medium in the SAM measurements is located between both glass substrates, is represented in light grey color. Bonded areas, in which there is no gap between both substrates, are shown in black dark color. Inhomogeneous grey-scale color would thus indicate different bonded degrees. As shown in the SAM micrographs in Fig. 3, the darkest and homogeneous irradiated blocks, and thus, those with the best bond quality are blocks A, C, and E, which were irradiated at 2 mm/s-14 passes, 2 mm/s-20 passes and 10 mm/s-40 passes, respectively. These three blocks A, C, and E show the highest blue emission intensity in Fig. 2b, too.

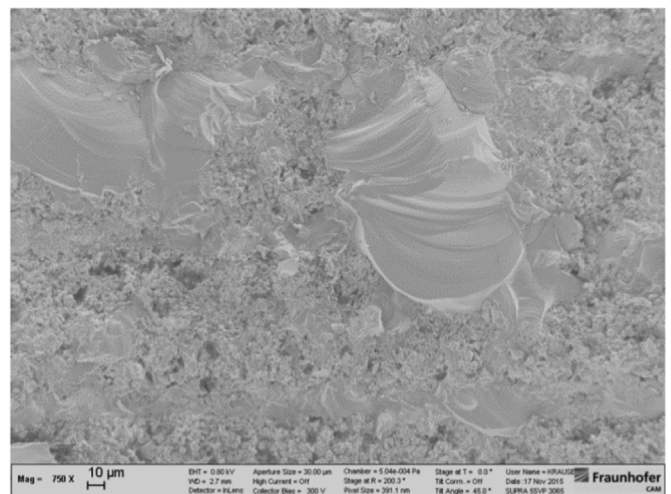
Following the laser parameters of the block C (2 mm/s, 20 Passes) further silica glass and sapphire substrates were joined using the BTS-3 SiO₂ glass thin film, by irradiating a 5×5 mm² square pattern of horizontal lines separated by 100 μm. The width of the irradiated wobble lines are about 200 μm, so a separation of 100 μm ensures the overlap of the irradiated lines. The joined substrates were mechanically separated and SEM micrographs were recorded from the glass sealant side of both, sapphire and silica glass substrates. Fig. 4 displays the SEM micrographs of the interface in the sapphire substrate (the initially coated side, cf. Fig. 1). Fig. 4a shows one of the corners of the irradiated square. The overlap between two irradiated lines is clearly observed, as well as their re-solidified microstructure. Some interesting features can be discerned in this interface, like seashell-like structures (Fig. 4b) and well-defined terraces (Fig. 4c).

The interface in the silica glass substrate was also investigated by SEM and is displayed in Fig. 5. The irradiated lines are also clearly discerned in Fig. 5a. The seashell-like structures are also present in this interface. Moreover, a granular structure is also observed in Fig. 5b with round particles around 500 μm.

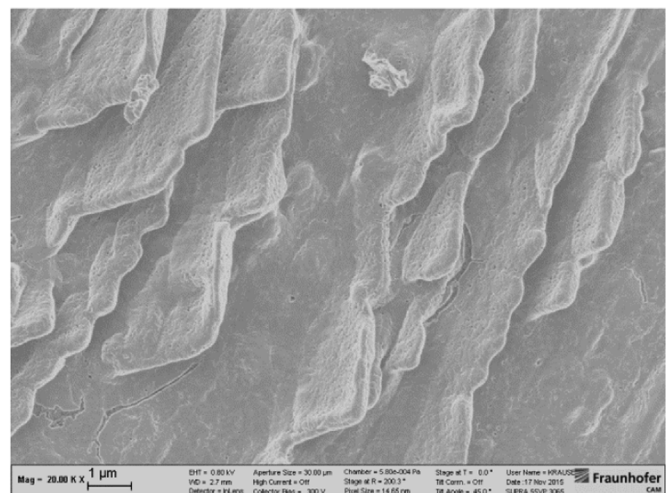
In order to go deeper in the composition of the glass solder, XRD diffractograms (not shown) were recorded at both interfaces. However, no diffraction peaks were discerned in any of the surfaces investigated. A further welded pattern was irradiated, consisting in two blocks of 20 horizontal lines. The first block was irradiated at 2 mm/s and 20 passes/line (process C), while the second block was performed at 5 mm/s and 20 passes/line as well (process D), in order to investigate the influence of the scan speed. Fig. 6 displays the cross-section of this sample, showing the gap between the two irradiated blocks. In the gap region, the pristine deposited thin film is observed. Contamination debris is also observed in this gap. At the right side of this micrograph the beginning of the block at 2 mm/s scan speed is discerned, while at the left side, begins the block at 5 mm/s. Each irradiated line forms a bump in the direction of the laser impingement (from the silica substrate). The height of the bumps is significant larger in those bumps



a



b

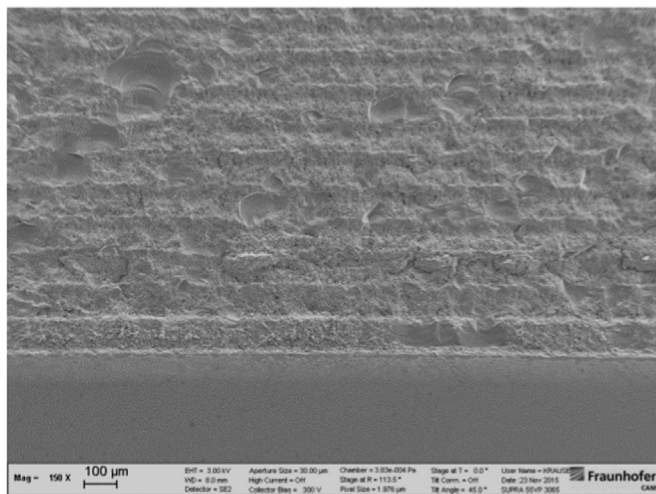


c

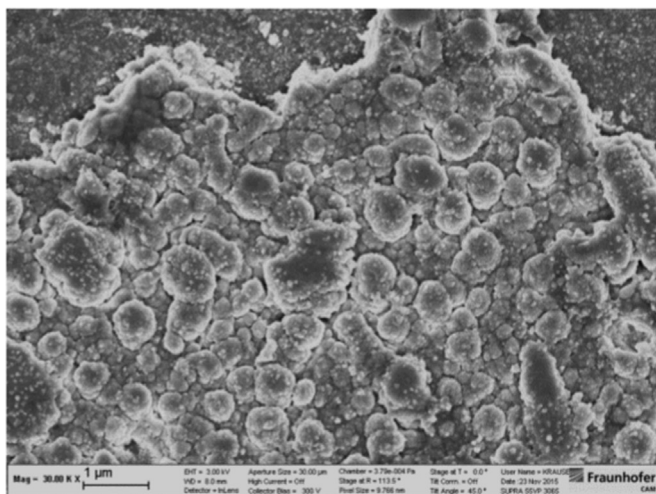
Fig. 4. SEM micrographs of the glass solder interface in the sapphire substrate, using parameter set C (Table 1). (a) Close-up to one corner of the irradiated square, (b) seashell-like structure and (c) well-defined terraces.

at 2 mm/s in comparison with those at 5 mm/s.

Fig. 7a displays a closer look of the bumps developed at 2 mm/s. It is observed that the damage of the bond interface in contact with the upper silica substrate gives rise to cracks and holes. The intermediate



a



b

Fig. 5. SEM micrographs of the glass solder interface in the silica glass substrate using the parameters C (Table 1): (a) One side of the irradiated square and (b) round particles observed at higher magnification.

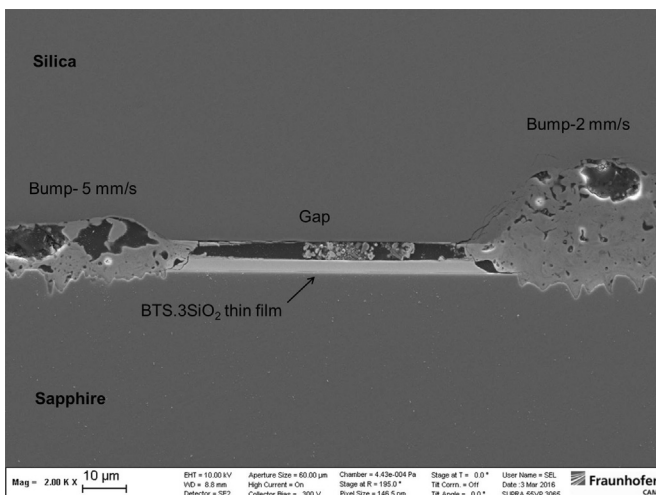
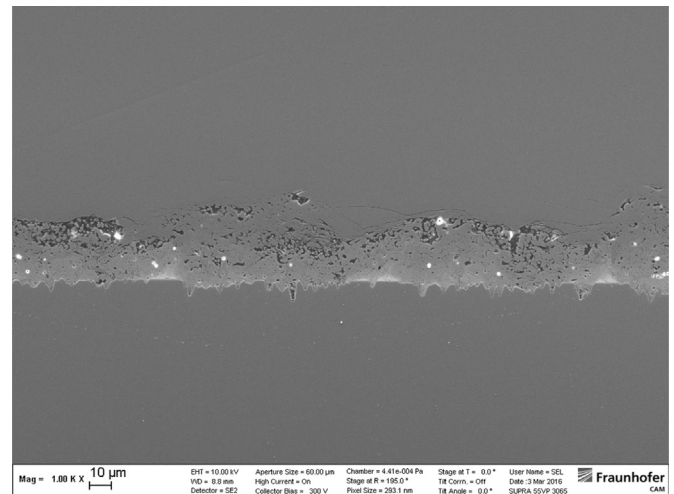
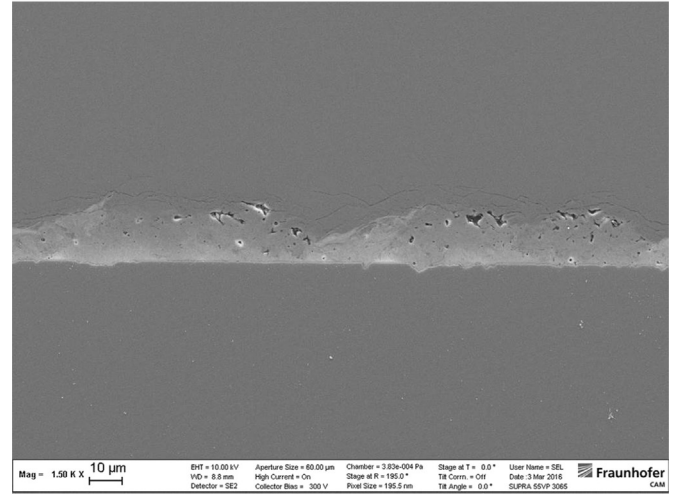


Fig. 6. SEM micrograph of the cross-section of the joined sample through two blocks of lines at scan speeds 5 mm/s and at 2 mm/s.



a



b

Fig. 7. SEM micrographs of the cross-section of the joined sample through two blocks of lines at scan speeds 2 mm/s (a) and at 5 mm/s (b).

thin film was melted, occupying the ablated space. The bottom sapphire substrate was also damaged at the interface, but to a much lower degree than the silica substrate. A small portion of the molten thin film occupies the ablated areas in the sapphire. At 5 mm/s the situation is similar, but the damage to the silica substrate is significantly lower, as well as the formation of cracks and holes is minimized, showing a smoother bond interface (Fig. 7b).

EDX elemental mappings were recorded in both blocks. Fig. 8 displays the one in the 2 mm/s block. Aluminum is concentrated in the sapphire substrate. However, aluminum is also present at the bottom of the formed bumps. The upper part of the sapphire substrate coinciding with the interface is ablated. Silicon is present in the silica substrate as well as in the thin film (BTS-3 SiO₂). Ba and Ti are concentrated in the thin film in an inhomogeneous distribution. The highest concentration was found near the interface with the sapphire substrate (initially coated, Fig. 1), while in the direction of the silica substrate their concentration is lower. Similar results are found when the scan speed is higher, as shown in the EDX elemental mappings in Fig. 9 for the 5 mm/s block. It is important to highlight that the Al incorporation in the thin film is much lower in this block in comparison with the one at 2 mm/s (Al-map in Fig. 8). In this block Al is practically limited to the sapphire substrate.

Besides elemental maps, EDX area analyses were performed in both blocks as well. The scanned areas are displayed in Fig. 10. For the block

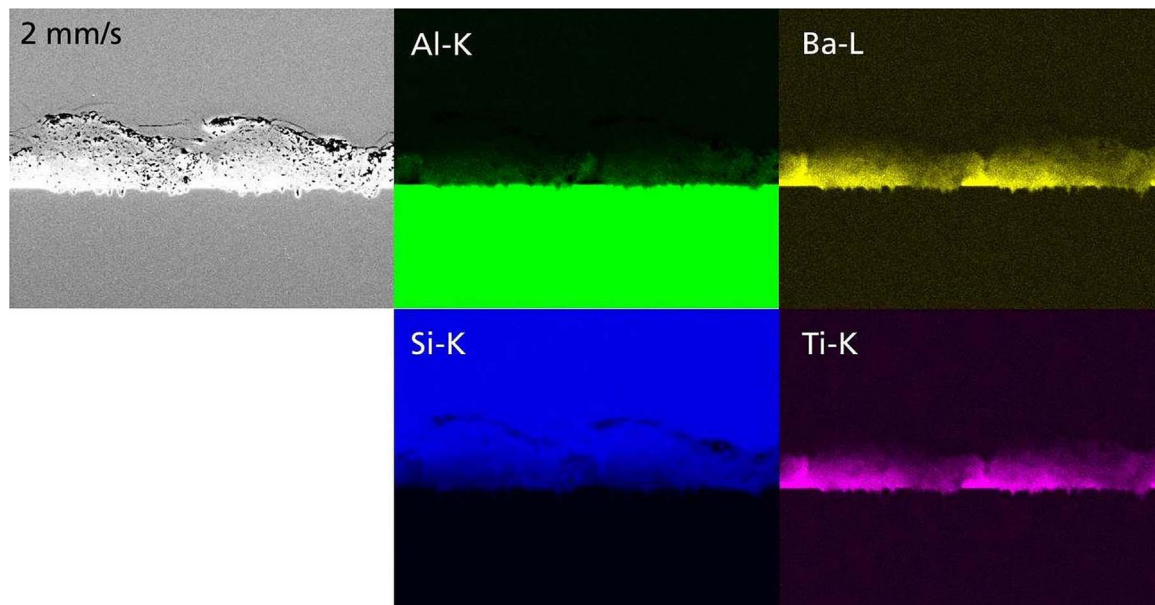


Fig. 8. EDX elemental mapping (Al, Ba, Si, and Ti) of the indicated area in the SEM micrograph at top left, corresponding to the block irradiated at 2 mm/s.

laser bonded at 2 mm/s, three different areas were analyzed. For comparison, the EDX point analysis in the unprocessed thin film in the gap between both blocks is also shown. The results are shown in Table 2. The unprocessed BTS-3 SiO₂ thin film shown in the gap in Fig. 10a presents a Ba:Ti:Si ratio of 1.62:1:3.88, which does not match exactly with that of the nominal BTS-3 SiO₂ glass composition, 2:1:5. The analyzed areas in the blocks show an inhomogeneous distribution of the elements, in good agreement with the elemental mappings in Figs. 8 and 9.

A cross-section sample of the joined sample C was prepared for TEM inspection. The glass solder presents a granular structure with features of some tens of nanometers (Fig. 11). Some of these features present darker points of some nanometers, especially observed at the edge of the solder in Fig. 11. Fig. 12 shows two views of the glass solder in the interface with the silica glass substrate. Round particles of different sizes, ranging between 10 and 40 nm are observed in the glass solder in the border with the silica glass substrate forming a row along it. It is necessary to underline that those particles were not found in the

interface with the sapphire substrate. An EDX line scan was performed across one of these particles (Fig. 13). It was possible to discern brighter regions within the round particles, of only some nanometers in size. The EDX analysis reveals that the round particles are enriched in Si, while the rest of the elements remain at low concentrations (Fig. 13).

In order to study the influence of the different CTE of silica glass and sapphire on the bond strength of the joint, one of the laser-bonded samples was heat treated at 100 °C for 30 min. As temperature reference, we considered the work in ref. [41], in which the range of working temperature in microfluidic systems is, depending on the system, below 100 °C. The CTEs of silica and sapphire in the range 25–200 °C are $0.5 \cdot 10^{-6} \text{ K}^{-1}$ and $6.6 \cdot 10^{-6} \text{ K}^{-1}$ (c-axis), respectively. Despite this large CTE difference and the large lateral extent of the bonded area ($5 \times 5 \text{ mm}^2$), the joined sample is stable and both substrates remain joined after the treatment. As a proof of that, the Fig. 14 shows the SAM micrograph of the heat treated sample. Non-bonded areas or delaminations were not observed.

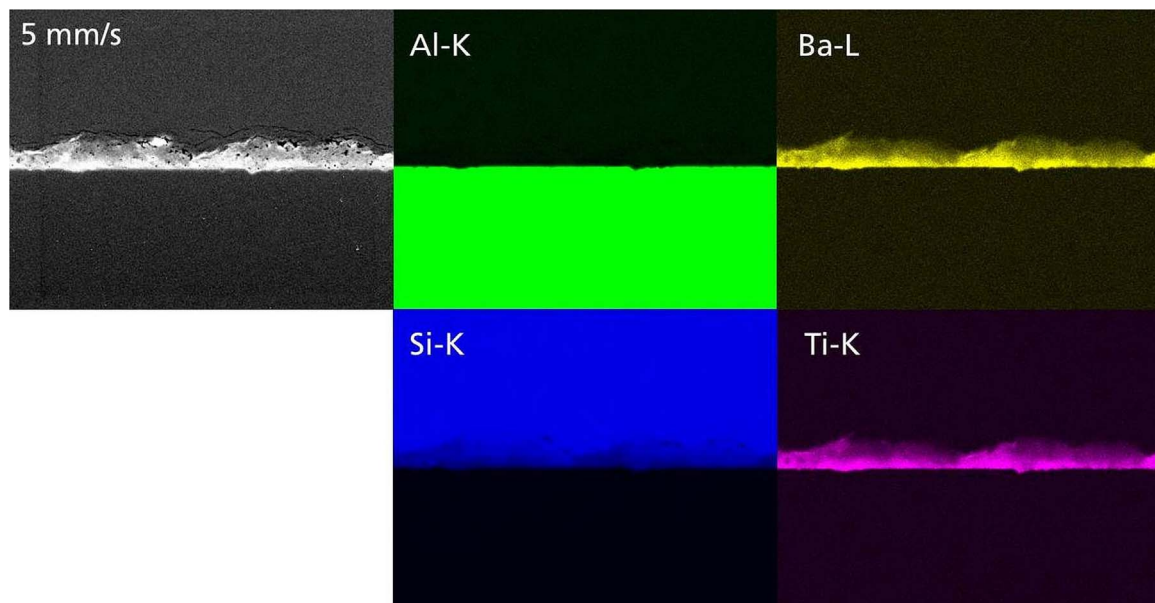


Fig. 9. EDX elemental mapping (Al, Ba, Si, and Ti) of the indicated area in the SEM micrograph at top left, corresponding to the block irradiated at 5 mm/s.

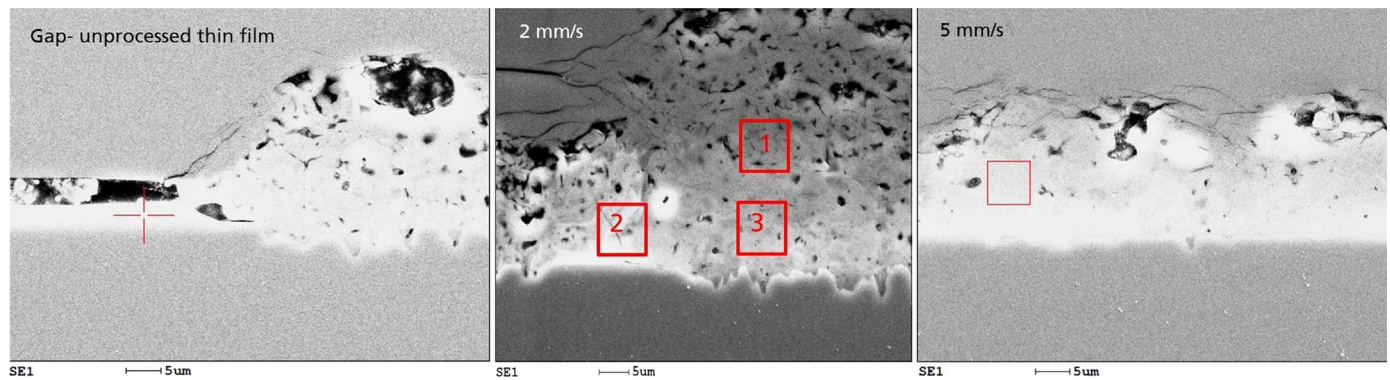


Fig. 10. Cross-section micrographs in three different positions of the bonded sample: in the gap between both blocks, showing the original thin film (left), bump of the block at 2 mm/s (middle) and bump of the block at 5 mm/s (right).

Table 2

Atom % by element of the scanned areas marked in Fig. 10.

Area/Atom-line	Al-K	Si-K	Ba-L	Ti-K	Ba:Ti:Si
Gap	21.33	46.99	19.58	12.11	1.62 : 1 : 3.88
2 mm/s-1	7.02	90.24	1.47	1.28	1.15 : 1 : 70.5
2 mm/s-2	21.08	64.07	8.19	6.55	1.25 : 1 : 9.88
2 mm/s-3	21.60	69.78	4.79	3.83	1.25 : 1 : 18.22
5 mm/s	6.00	85.65	4.58	3.78	1.21 : 1 : 22.66

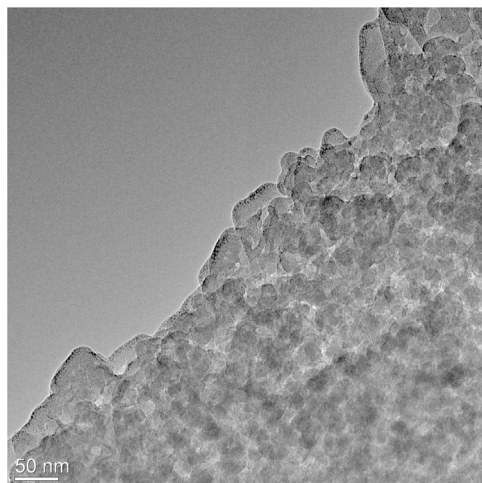


Fig. 11. TEM micrograph of the glass solder in sample C.

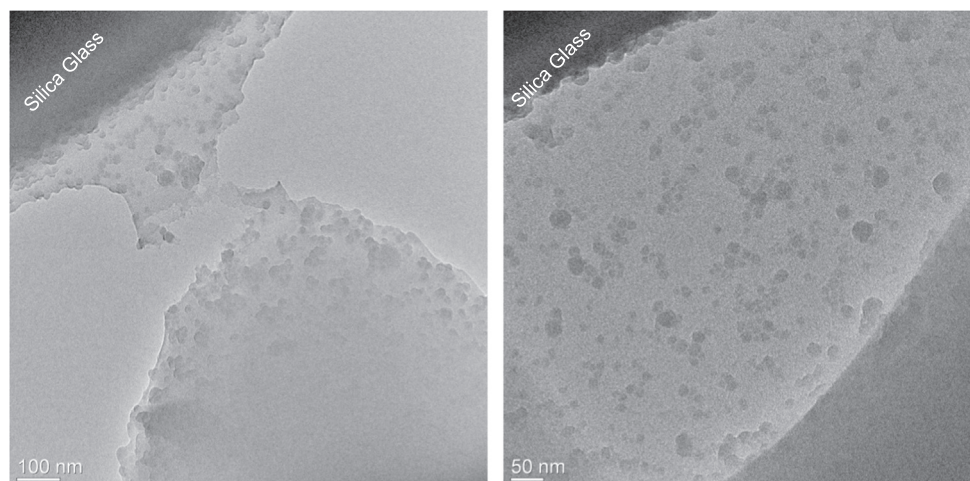


Fig. 12. TEM micrographs at different magnifications of the glass solder in sample C at the border with the silica glass substrate.

4. Discussion

In our previous work [30,31] it was stated that the upper substrate, that is, the substrate through which the laser reaches the interface, becomes damaged at the interface (the surface remains intact) during the laser welding process and thus, its components are incorporated into the glass solder composition. Having this behavior in mind, the silica glass was selected to be the upper substrate, instead of the sapphire one. In this way, the incorporation of the silica glass will lead to an enrichment of SiO_2 in the glass solder. Otherwise, the Al incorporation from the sapphire substrate would inhibit the crystallization of BTS upon laser irradiation, as previously shown in [31].

From all the laser parameters tested, the block C shows the best bond-interface quality in the SAM micrograph (Fig. 3), together with the highest luminescence intensity under UV excitation (Fig. 2b). This result points out that a higher pulse overlap (slow scan speed) and a higher number of passes enhance the quality of the bond and support the crystallization of BTS, and hence leading to the observed high luminescence intensity. The laser parameters of the block C were selected to develop additional samples and to carry out their structural characterization.

The melt of the intermediate thin film is clearly observed in the microstructure of the separated substrates (Figs. 4 and 5). It is also observed how the overlapping between irradiated lines creates a fully bonded area. The observed terraces in Fig. 4c are similar to those reported in [42] for BTS thin films grown on a-plane sapphire at 1250 °C. This structure indicates a step-flow growth. After re-solidification the glass solder formed characteristic features in form of round particles and seashell-like forms. The XRD diffractograms (not shown)

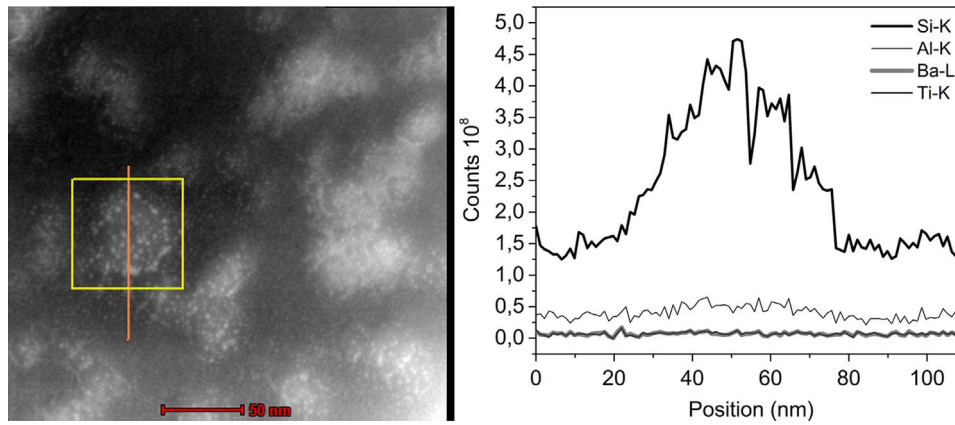


Fig. 13. Left: STEM micrograph of the glass solder in sample C with the location of the line scan indicated as an orange line. Right: EDX line scan along the orange line indicated in the micrograph.

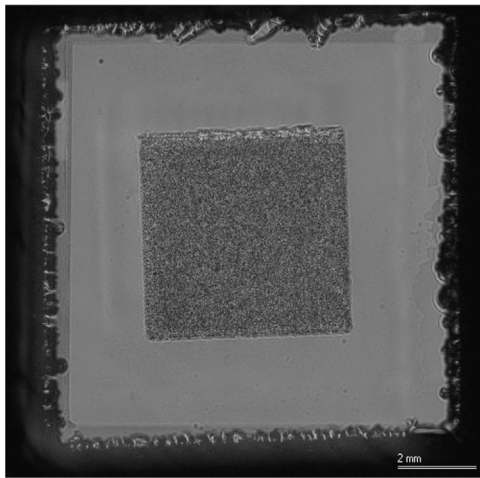


Fig. 14. SAM micrograph of the joined sample consisting in Saphir/BTS.3SiO₂/Silica glass after 100 °C annealing for 30 min.

in these interfaces reveal that they are amorphous structures.

The laser beam passes through the fused silica substrate and reaches the BTS-3 SiO₂ absorbing thin film. With nanosecond pulsed lasers at high peak intensities, the material is changing its absorption characteristics [43], e.g., by the reduction of polyvalent elements like titanium [44] in the fresnoite composition. This photodarkening leads to increased absorption towards the temporal end of each individual pulse and consequently ablation. In fact, cracks and micro-holes are observed in the interior of the substrates as a consequence of this ablation (Fig. 7) and fused silica from the substrate is incorporated into the molten glass solder. Cvecek [45] reported the most common defects in glass/glass interfaces bonded through laser welding. It was stated that the cracks at the beginning of the weld are generated when the laser hits the cold material, leading to a thermal shock. The influence of the scan speed is displayed in Fig. 7. At the lower speed of 2 mm/s, the pulse overlap is larger, leading to a larger heat affected zone (HAZ) (higher bumps in Fig. 6) in comparison with a higher speed of 5 mm/s.

The composition of the BTS-3 SiO₂ thin film in the gap (Fig. 10a) does not match exactly with that of the nominal BTS-3 SiO₂ glass composition, 2:1:5. Si-deficient phases in fresnoite thin films deposited by PLD were reported and attributed to different chemical stability of the oxides to hinder the Si to be ablated in stoichiometric ratio [42]. This explains a lower Si content in the thin film in comparison with the original BTS-3 SiO₂ glass. Surprisingly, the unprocessed thin film contains a high Al content (Table 2). It is possible, that Al comes from the inter-diffusion from the sapphire substrate at the thin film during PLD. Similar behavior was observed in ZnO thin films grown on sapphire [35].

The temperature reached by the laser interaction can be of the order of thousands of Kelvin, which favors the diffusion of the atoms and the melting of the material. Senz et al. [46] studied the diffusion processes and the BTS crystallization on a SiO₂ thin film deposited on a BaTiO₃ substrate. Due to the high viscosity of the SiO₂ glass substrate, the diffusion of Ba and Ti in the interior of the substrate was reported to be limited. This could explain the lower concentration of Ba and Ti on the top of the bump in comparison with those in the areas near to the initially coated sapphire substrate (Fig. 10 and Table 2). Thus, the thin film exhibits an inhomogeneous composition and the BTS crystallization and the observed luminescence will not take place in a homogeneous way in the whole irradiated bump. The fluorescence observed in the bonded samples agrees with the crystallization of fresnoite between both fused silica substrates. The excitation at 254 nm corresponds to the O²⁻–Ti⁴⁺ charge transfer state (CTS) from the 2p orbital of O²⁻ to the 4d orbital of Ti⁴⁺. This transition is common for titanates or Ti-activated compounds [47]. Aluminum concentration is higher in those areas near to the sapphire substrate, that is, areas 2 and 3, while it is lower in area 1, situated in the ablated silica substrate. This points out the damage of the bond interface in contact with the bottom sapphire substrate upon the laser process, although much moderate than in the silica substrate. The enhanced silicon concentration comes from the silica content in the thin film and also from the ablation of the silica substrate at the interface. Thus, in area 1 near the silica substrate its content is the highest. In the bump at 5 mm/s, Al content is lower than that at 2 mm/s, according with a higher radiation damage at lower scan speeds. At this point, the following features of the laser welding process in this system can be summarized:

1. The PLD deposited thin film contains Al from the substrate from the PLD process.
2. Both substrates become damaged at the interface upon the laser welding process. However, the damage of bond interface with the silica substrate (facing the impinging laser beam) is more severe than that of the underlying sapphire (far side).
3. The bumps show a gradual Ba and Ti distribution, in which both elements are more concentrated in the sapphire interface and diffuse towards the silica substrate.
4. The damage of the bond interface is more pronounced at 2 mm/s than at 5 mm/s, due to a higher pulse overlap.

From the microstructure revealed by TEM, similar darker features in Fig. 11 were found in a sapphire/BTS/sapphire bond after heat treatment [31], which were attributed to BTS crystals. Additionally, the formation of Si-enriched particles in the interface in contact with the silica glass substrate was confirmed (Fig. 13). Höche et al. [48] reported the crystallization of a BTS glass with an excess of SiO₂ by

electrochemically induced nucleation. It was stated that fresnoite dendrites crystallize, while in the interdendritic areas, the pseudo-binary eutectic fresnoite-silica is formed lamellarly with alternating layers of crystalline fresnoite and glassy silica. The microstructure of round particles in Fig. 13 resembles the one reported by Takahashi et al. [49], which were attributed to SiO_2 particles. It was also reported a region with a decrease of Sr in $\text{Sr}_2\text{TiSi}_2\text{O}_8$ fresnoite-type glass-ceramics. In fact, they confirmed a SrO-rich phase, which corresponds to the fresnoite stoichiometry and a residual SiO_2 -rich phase [49]. The nanostructures are slightly different of that reported in other works, presumable attributed to differences in glass compositions (i.e. $34.8\text{SrO}-17.4\text{TiO}_2-47.8\text{SiO}_2$ by Wisniewski et al. [50]) and the differences in the processing. In Ref. [50] Wisniewski et al. reported two fresnoitic glasses with excess of SiO_2 , of composition $\text{Sr}_2\text{TiSi}_{2.45}\text{O}_{8.9}$ and $\text{Sr}_2\text{TiSi}_{2.75}\text{O}_{8.5}$. The excess SiO_2 occurs in the form of a SiO_2 -enriched residual glassy phase after crystallization.

There are several reasons which support the crystallization in the interface with the silica glass and not (or poorly) in the interface with the sapphire: (1) Fused silica presents a low thermal conductivity. It is more than one order of magnitude lower than that of sapphire, 1.4 versus 42 W/m K at 25 °C, respectively, and very similar to that of fresnoite (0.71–1.83 W/Km, depending on the orientation [51]). Thus, fused silica would dissipate poorly the heat originating from laser irradiation, leading to self-heating and favoring the heat accumulation in its interface with the glass thin film. On the other hand, sapphire acts as a heat spreader material and the heat produced by the laser irradiation in its interface with the thin film is dissipated from the stack through the sapphire substrate. The low thermal conductivity in silica glass explains also the higher damage upon irradiation in comparison with the sapphire substrate. (2) Ti^{4+} is a network former in four-fold coordination. With increasing concentration of SiO_2 (network former), Ti^{4+} will be displaced from the network former sites to Ti^{4+} in five- and six-fold coordination, acting as a network modifier [52] and thus, favoring the crystallization of BTS in the interface with the silica glass. (3) Al incorporation into the sealant hinders BTS crystallization, as already reported in sapphire/BTS/sapphire joints [31]. This behavior explains the absence of diffraction peaks in the XRD, since the studied interface corresponds to the middle of the glass solder, and the crystals are precipitated in a layer near the silica glass substrate (Fig. 12).

Despite the large CTE difference between both substrates, the substrates remain bonded even after heat treatment at 100 °C during 30 min. Incorporation of Si and Al form the substrates into the glass solder supports a lower CTE difference between the glass solder and the bonding partners mediated by an intermediate slab.

Finally, this article is mainly focused on the feasibility study of the laser welding of two dissimilar materials like sapphire and silica glass by employing a ns pulsed laser, and on the structural characterization of the bond interface. The mechanical characterization and evaluation of the mechanical strength is part of a future work. However, from the SAM micrographs (Fig. 3), the parameters used in block C (2 mm/s and 20 passes at 35 kHz and 8.25 J/cm² fluence) are expected to yield the highest bonding strength.

5. Conclusions

The feasibility study of the laser welding of fused silica glass with sapphire using a nanosecond pulsed laser was put forward, as an alternative to ultra-short pulse lasers.

A SiO_2 -enriched fresnoite glass thin film acts as intermediate thin film sealant. Laser parameters, particularly scan speed and number of passes per line, were evaluated, in terms of substrate damage at the interface, achieved bonded area and luminescence of the bonded area under UV irradiation. A higher pulse overlap (slow scan speed) and a higher number of passes enhance the bond area and support the crystallization of BTS, leading to high luminescence intensity. Laser

irradiation triggers the crystallization of fresnoite, together with the formation of SiO_2 particles, mainly in the bond interface in contact with the silica substrate, rather than with the sapphire substrate. The technological importance of the developed joints lies, in the blue luminescence of the stack under UV excitation, and in the stability of both substrates, even after heat treatment at 100 °C for 30 min, despite the large CTE difference between them. The developed joints represent a technological advance and open new possibilities in the fields of MEMS packaging, microfluidics, etc.

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